Cluster Innovation Centre

University of Delhi
End Semester Examination December 2025

Name of the Course: B.Tech (IT & MI)
Semester: V
Paper Title: **Genes to Genomes**
Paper code: **3124000019**
Maximum marks: **60** Time duration: 2 hours
Instructions: **Attempt any four questions. All questions carry equal marks.**

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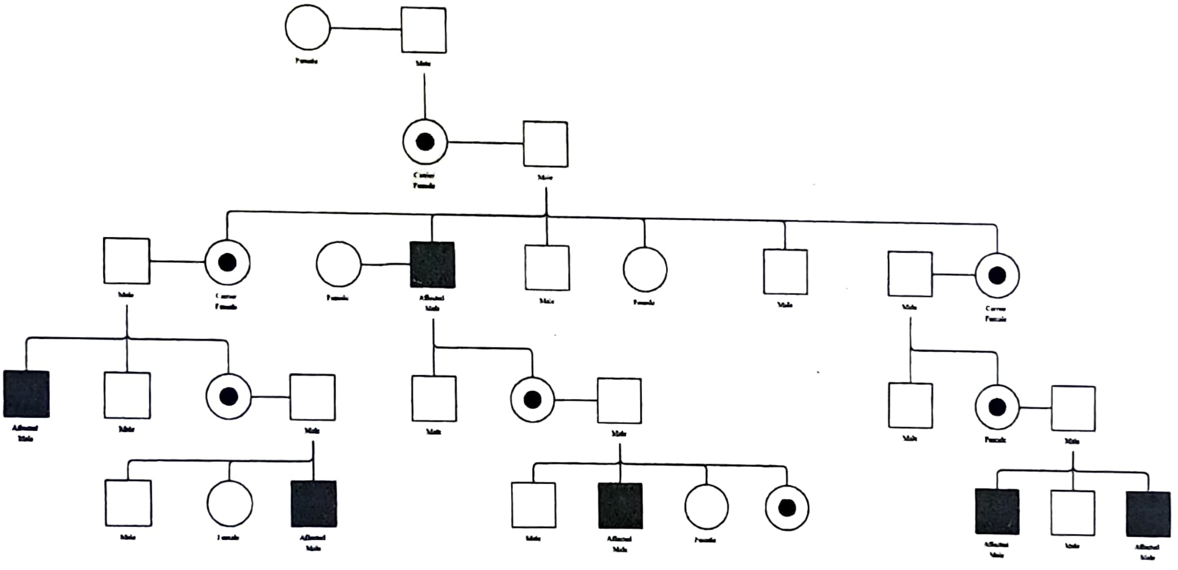
1. Explain any three: Sickle Cell Anemia, Paracentric Inversion, Histone Octamer, Recessive Epistasis (5+5+5=15)

2. Deduce the following Pedigree Chart and comment on the genetic nature of the condition depicted in this chart (15)



3. Explain the following ratios with Punnett square: A)**12:3:1,** B) **2:1,** C) **9:3:4** (5+5+5=15)

4. A. In Labrador Retrievers, coat color is controlled by two genes. The B gene controls black (B) vs. brown (bb) color, while the E gene controls yellow (ee) color. The recessive yellow genotype is epistatic to the B gene which is otherwise dominant. What genotypic and phenotypic ratio will result in mating of two heterozygotes Labradors (BbEe)? (10)
B. In humans, red-green colorblindness is a recessive trait found on the X chromosome. Normal vision is dominant to color-blindness. Show the possible children of the following couples: A male with red-green colorblindness and a heterozygous normal vision woman. (5)

5. Determine the map distance and orientation among the genes from the following three point cross

(15)

ABC	349
Abc	114
AbC	5
aBC	116
abC	128
aBc	4
ABc	124
abc	360